D4.3 Climate change

NAME	DATE	TOTAL
		_____ / 36

About 35 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 STATE two human sources each of carbon dioxide and of methane. [D4.3.1] [2]

2 DEFINE tipping point. [D4.3.3] [1]

3 STATE the type of ice that emperor penguins breed on. [D4.3.4] [1]

4 LIST three approaches to carbon sequestration. [D4.3.8] [3]

Section B: Understanding

13 MARKS

5 EXPLAIN how thawing permafrost is an example of positive feedback. [D4.3.2] [3]

6 EXPLAIN how warmer surface water can reduce the productivity of an ocean. [D4.3.5] [3]

7 EXPLAIN why species living near the tops of mountains are at particular risk. [D4.3.6] [2]

8 **DISTINGUISH** between ocean acidification and coral bleaching. [D4.3.7] [3]

ACIDIFICATION	BLEACHING

9 **EXPLAIN** why restoring a peat wetland helps to store carbon. [D4.3.8] [2]

Section C: Working with data

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

10 [D4.3.6]

The mean elevation of three bird species on one mountain was recorded in two surveys, 40 years apart.

SPECIES	FIRST SURVEY / M	SECOND SURVEY / M
A	1200	1420
B	1800	2010
C (TOP OF MOUNTAIN)	2600	Not found

a **CALCULATE** the mean rate of upslope movement of species A, in metres per decade. [1]

b **EXPLAIN** the results for species A and B. [2]

c **SUGGEST** two explanations for the result for species C. [2]

11 [D4.3.7]

The rate of calcification in a coral was measured at different values of pH.

PH	8.2	8.0	7.8	7.6
RELATIVE CALCIFICATION / %	100	85	62	38

a **DESCRIBE** the relationship. [1]

b **EXPLAIN** how a rise in atmospheric carbon dioxide could produce these effects in the sea. [3]

c **SUGGEST** one effect of lower calcification on other reef species. [1]

Section D: Longer answer

6 MARKS

12 **DISCUSS** the effects of climate change on polar and marine ecosystems. [D4.3.4, D4.3.5, D4.3.7] [6]